

Mark Jennings

Robotics Engineer | markjennings97@gmail.com | <https://makr.org>

Work Experience	Education
Elevate Robotics Robotics Engineer, September 2024 – Present - Prototyped a series of lightweight, high-payload robotic manipulators with passive gravity balancing - Integrated payload detection and halting functionality with load cell and patented fail-safe braking system - Led mechanical analysis and technical writing for NASA SBIR contract, successfully secured next phase for \$850k	MS Mechanical Engineering UT Austin, 2019 – 2021 - Thesis: Manipulator Control in Collaborative Assembly - Funded by DOE
Los Alamos National Laboratory R&D Engineer, October 2021 – August 2024 - Overhauled nuclear glovebox with the first autonomous robotic arm in US plutonium part production - Developed control software, operating procedures, and tooling for multiple nuclear manufacturing processes - Coordinated division intern program	BS Mechanical Engineering UT Austin, 2015 – 2019
Nuclear and Applied Robotics Group Graduate Research Assistant, August 2019 – August 2021 - Developed a C++ package to augment assembly tasks with a collaborative robot via force-sensing and admittance control - Refactored the software stack for a custom mobile manipulator, leveraging open-source packages	Skills
Sandia National Laboratory R&D Intern, Summer 2019 Led 1st place intern team to design additively manufactured parts and test them in lab-simulated launch conditions	Mechanical: - Design: CAD (Onshape, SolidWorks, Creo), GD&T, DFMA, FEA - Fabrication: CNC/manual machining, additive manufacturing, casting, extrusions, sheet metal, welding - Electric actuation: motors, drivers, reduction/transmission, brakes, bearings, encoders, control/filtering
Apptronik Mechanical Engineering Intern, Summer 2018 - Derived forward kinematic equations for company's first humanoid bipedal robot, Draco - Updated actuator testbed product to achieve higher payloads and modeled heat flow to maximize torque density	Software: - Languages: C/C++, Python - Algorithms: mobile robot localization/mapping/navigation, manipulator motion planning - Other: Linux, Git, ROS2, various sims
Rehabilitation and Neuromuscular (ReNeu) Robotics Lab Undergraduate Research Assistant, May 2016 – May 2019 Designed and fabricated parts for rehabilitation exoskeleton robots and prosthetics	Outreach
	Los Alamos FIRST Tech Challenge Mentor/Coach, 2022 – 2024 Taught middle schoolers STEM, problem-solving, and teamwork
	UT Robotics & Automation Society Mentor/Officer, 2015 – 2019 Mentored first-year robotics teams and held several leadership roles